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SMART ANTENNA POSITIONING & ALIGNMENT SYSTEM WITH CLOUD MONITORING

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for the Course
of*

ECA1206-ANTENNA WAVE AND PROPAGATION

to the award of the degree of

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Smart Antenna Positioning & Alignment System with Cloud Monitoring” has been carried out by Santhosh D (192412209), A.K. Likith Charan (192412305), Sanjay Anand (192412036), Balakumar (192412221) and Mahendran (192412159) under the supervision of Dr. Saravana Kumar R and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Santhosh D, A.K. Likith Charan, Sanjay Anand, Balakumar and Mahendran of the Department of Electronics and Communication Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Smart Antenna Positioning & Alignment System with Cloud Monitoring” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. %
Santhosh D 192412209	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_20_%]
A.K. Likith Charan 192412305	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_20%]
Sanjay Anand 192412036	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_20_%]
Balakumar 192412221	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_20_%]
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Total					100%

ABSTRACT

Directional antennas used in satellite communication, long-range Wi-Fi links, IoT sensor networks and drone/UAV ground stations must be pointed with precision to maintain reliable signal strength, yet manual alignment is slow, error-prone and cannot adapt to gradual mechanical or environmental drift. This project addresses that engineering problem by developing a Smart Antenna Positioning and Alignment System with cloud monitoring, capable of automatically orienting an antenna to the direction of best signal strength while eliminating the effort and inaccuracy of manual adjustment. The proposed solution uses an ESP32 microcontroller as the central processing unit, driving two servo motors to achieve independent azimuth (0–180°) and elevation (0–180°) control through 50 Hz PWM signals with 1° angular resolution. A Light Dependent Resistor (LDR) array performs differential optical tracking to detect the direction of strongest signal/light source, while a DHT11 sensor monitors ambient temperature (0–50°C, $\pm 2^\circ\text{C}$) and humidity to account for environmental effects on alignment. A closed-loop state-machine algorithm continuously compares the current orientation against calibrated reference angles to eliminate overshoot and mechanical backlash, and a local HD44780 LCD display shows live positional angles and environmental readings. The system additionally transmits telemetry to a cloud-based dashboard, giving technicians real-time dial gauges, historical logs, threshold alerts and manual override sliders accessible from any smartphone or web browser. The design methodology followed a systems-engineering approach spanning mechanical actuation, environmental sensing, embedded processing and IoT communication, integrated through a regulated dual-rail (3.3 V / 5.0 V) power distribution circuit for reliable operation. The prototype demonstrates stable dual-axis positioning, real-time environmental feedback and functional remote monitoring through the cloud interface. The principal conclusion is that combining low-cost embedded control with closed-loop sensing and cloud connectivity significantly reduces manual effort and alignment error while improving communication reliability, making the system suitable for satellite dish alignment, rural point-to-point Wi-Fi links, wireless sensor networks and drone ground stations.

Keywords: Antenna positioning, dual-axis alignment, ESP32, closed-loop feedback, cloud telemetry, IoT monitoring.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
iii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	1
2	CHAPTER 2: Literature Review	3
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	5
4	CHAPTER 4: Implementation, Testing & Results, Project Management	10
5	CHAPTER 5: Conclusion & Reflection	13
	References	14
	Appendices	15

Note: page numbers above are indicative — update the field/table once the document is finalised in Word (References → Update Table, or manually adjust).

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1.1	Antenna alignment concept	1
3.1	System block diagram – Smart Antenna Positioning System	8

LIST OF TABLES

Table No.	Table Name	Page No.
3.1	Stakeholder / User Requirements	5
3.2	Functional & Non-Functional Requirements	5
3.3	Design Specifications	6
3.4	Hardware Component Specifications	9
4.1	Test Plan & Verification	11
4.2	Comparison with Existing Solutions & Achievement of Objectives	12

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
PWM	Pulse Width Modulation	—
ADC	Analog-to-Digital Converter	bit
LDR	Light Dependent Resistor	—
IoT	Internet of Things	—
GUI	Graphical User Interface	—
Azi	Azimuth angle	°
Ele	Elevation angle	°
RH	Relative Humidity	%
SRAM	Static Random Access Memory	KB
BLE	Bluetooth Low Energy	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Directional antennas are widely used in satellite communication, long-range Wi-Fi links, wireless sensor networks (WSN) and drone/UAV ground stations. The quality of the received or transmitted signal depends heavily on how precisely the antenna is pointed toward the source or target. Traditionally, antenna alignment has been carried out manually by technicians who adjust the azimuth (horizontal) and elevation (vertical) angles by hand while observing a signal-strength indicator. This process is slow, requires trained personnel, and is difficult to repeat consistently, especially in remote or hard-to-access installations such as rooftop dish mounts, rural relay towers or field-deployed sensor nodes. Advances in low-cost embedded microcontrollers (such as the ESP32), micro-servo actuators and IoT cloud platforms now make it feasible to automate this process and monitor it remotely.

1.2 Need for the Project

- Manual alignment is time-consuming and depends on the skill of the technician, leading to inconsistent results.
- Environmental factors such as wind, temperature and vibration gradually shift antenna orientation, but manual systems have no way of detecting or correcting this drift automatically.
- Remote or elevated installations (towers, rooftops, rural sites) are difficult and unsafe to access frequently for re-alignment.
- Field technicians, network operators and IoT/UAV ground-station operators are directly affected by poor alignment, which shows up as weak signal, dropped links and increased site visits.
- There is a clear engineering significance in combining precise dual-axis actuation, closed-loop feedback and cloud connectivity into a single automated, remotely monitorable system.

1.3 Problem Statement

To design and develop an automated dual-axis antenna positioning and alignment system that (i) drives azimuth and elevation servo actuators with fine (1°) angular resolution under real-time closed-loop feedback to eliminate overshoot and mechanical backlash, (ii) senses environmental conditions (light intensity, temperature and humidity) that influence signal quality and mechanical behaviour, (iii) reconciles the conflicting requirements of fast positioning response, low power consumption and mechanical rigidity under wind/vibration loading, and (iv) exposes live status and manual override control through a cloud-based mobile/web dashboard — while keeping the hardware low-cost and simple enough to deploy at remote or unmanned sites.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

[Describe how you identified relevant literature, standards, patents and commercial products — e.g. databases searched (IEEE Xplore, Google Scholar), search keywords used, inclusion/exclusion criteria, and the number of sources reviewed. This section must be completed with your own search process.]

2.2 Review of Existing Work

[This section requires a genuine literature survey and must be written and referenced by the team — it cannot be generated from the presentation alone. Structure it around: (a) manual/rotator-based antenna positioners, (b) satellite auto-tracking systems, (c) IoT-enabled antenna/solar-tracker positioning systems, (d) relevant commercial products. Critically discuss each category's approach, and cite every source in Chapter References using IEEE style.]

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual antenna alignment	Simple, low cost, no electronics required	Slow, imprecise, requires trained technician on-site	Motivates the need for automated dual-axis control
Fixed rotator controllers (non-IoT)	Accurate mechanical positioning	No remote monitoring; status only visible on-site	Motivates addition of cloud telemetry
[Add reviewed system / paper]	[...]	[...]	[...]

2.4 Research / Engineering Gap

[State clearly, based on 2.2–2.3, what remains unresolved in existing approaches — e.g. lack of combined closed-loop dual-axis control with low-cost environmental compensation and cloud-based manual override in a single low-cost platform.]

2.5 Proposed Contribution

The proposed system differs from purely mechanical rotator controllers by combining closed-loop dual-axis servo control, LDR-based differential optical tracking, environmental (temperature/humidity) compensation, and a cloud-connected mobile dashboard with manual override sliders — all on a single low-cost ESP32 platform, enabling both automatic alignment and remote human supervision from any location.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Field technician / installer	Ability to align the antenna without manual crank/adjustment; view live angle on local display.
Remote network operator	View real-time signal/environmental telemetry and override positioning from a phone or browser from any location.
System owner	Low power consumption suitable for solar/battery operation; durable outdoor mount.

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Independent azimuth and elevation control	0–180° sweep on each axis
Functional	Angular positioning resolution	1° step resolution
Functional	Environmental sensing	Temperature 0–50°C ±2°C, humidity, light intensity
Functional	Remote monitoring & override	Manual override sliders via cloud dashboard
Non-Functional	Sensor sampling rate	1 Hz background sampling
Non-Functional	Servo control signal	50 Hz PWM, 1.0–2.0 ms duty cycle
Non-Functional	Power supply stability	Regulated 3.3 V logic / 5.0 V motor rail, 2 A peak

3.2 Constraints & Applicable Standards

[Verify and cite the exact standards/codes your institution requires you to follow — the entries below are the categories typically relevant to this project and should be confirmed against actual test/compliance evidence before submission.]

Standard / Code	Requirement	How Applied in This Project
IEEE 802.11 (Wi-Fi)	Wireless communication interoperability	ESP32 Wi-Fi module used for cloud telemetry link
[Enclosure/ingress protection standard, e.g. IEC 60529]	Outdoor enclosure protection rating	[State the IP rating designed for / achieved]
[Institutional safety code]	Electrical safety of low-voltage DC systems	Regulated, fused 3.3 V / 5 V rails with thermal shutdown

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Azimuth range	0–180	°	High
Elevation range	0–180	°	High
Angular step resolution	1	°	High
Servo PWM frequency	50	Hz	High
Servo duty cycle range	1.0–2.0	ms	High
ADC resolution	12 (4096 steps)	bit	Medium
Sensor sampling rate	1	Hz	Medium
Temperature monitoring range	0–50 ± 2	°C	Medium
Motor rail current (peak)	up to 2	A	High

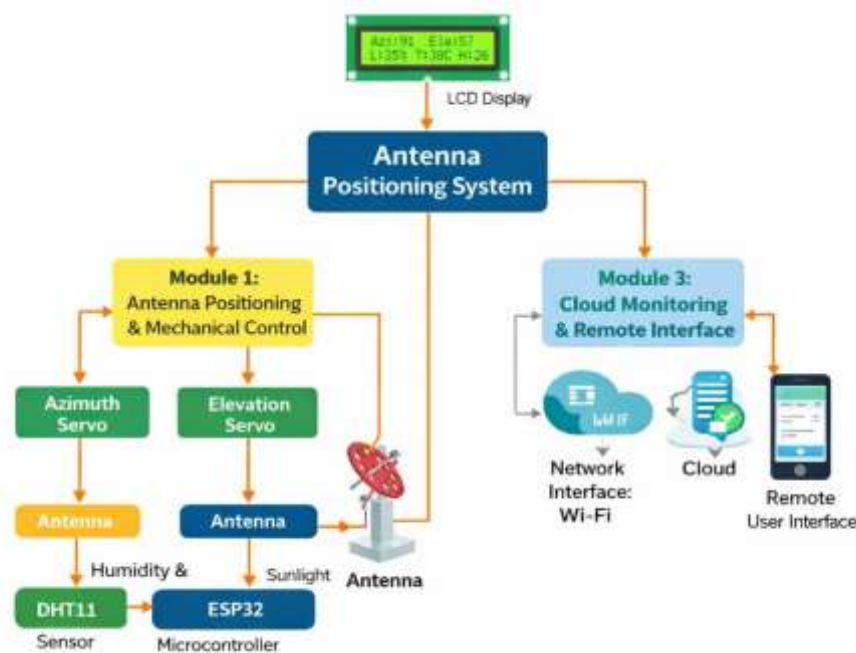


Figure 3.1: System block diagram – Smart Antenna Positioning System

The system architecture (Figure 3.1) is organised into five interfacing modules: Module 1 (Mechanical Control & Actuation) drives the azimuth and elevation servos from PWM commands generated by the ESP32; Module 2 (Sensing & Environmental Feedback) supplies LDR and DHT11 readings back to the control loop; Module 3 (Embedded Processing & Local Display) runs the closed-loop alignment algorithm and drives the LCD; Module 4 (Cloud Telemetry & Remote IoT GUI) publishes telemetry and receives override commands; and Module 5 (Power Management) supplies clean, regulated power to all subsystems. Each module communicates through the ESP32 core, which acts as the single point of coordination between sensing, actuation and cloud connectivity.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Actuator type	Stepper motor	No feedback needed for open-loop stepping	Higher cost, more complex driver circuitry	Servo motors selected — simpler PWM control and adequate for 0–180° range
Tracking sensor	Camera-based tracking	Higher precision, works day/night with IR	Higher cost, higher processing load	LDR array selected — low cost, sufficient for light-source differential tracking
Connectivity	GSM/cellular module	Works without local Wi-Fi	Recurring SIM cost, higher power draw	Built-in ESP32 Wi-Fi selected — lower cost and power for the intended deployment

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Energy efficiency of the control platform	ESP32 power draw compared against general-purpose PC/heavier MCU alternatives (Section: Advantages)	Selected ESP32 to allow future solar/battery-powered deployment
Ethics	Transparent disclosure of automated decisions to operators	Manual override sliders retained on the dashboard rather than fully autonomous control	Dual manual/automatic mode included so a human can always intervene
Health & Safety	Electrical safety of motor and logic rails	Thermal shutdown safeguard and decoupling capacitors evaluated against brownout/stall conditions	Dedicated regulated 3.3 V/5 V rails with thermal shutdown adopted
Security	Unauthorised remote access to the antenna control dashboard	Requirement for secure end-to-end access reviewed against open/unauthenticated dashboard designs	Access restricted to authenticated sessions on the mobile/web dashboard [confirm and document actual authentication mechanism used]

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Mechanical backlash causing angular drift	Medium	Medium	Medium	Closed-loop feedback continuously re-checks angle against reference	Low
Wi-Fi link loss interrupting telemetry	Medium	Low	Low	Local LCD continues showing live angles/readings independent of cloud link	Low
Motor stall / over-current damaging driver	Low	High	Medium	Thermal shutdown safeguard on 5 V motor rail	Low
Unauthorised access to control dashboard	Low	Medium	Low	[Confirm authentication method used]	[Assess after implementation]

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The prototype was developed following an iterative, module-by-module approach: mechanical actuation (Module 1) and sensing (Module 2) were built and bench-tested first, followed by embedded control and local display (Module 3), then cloud telemetry (Module 4), with power distribution (Module 5) integrated and validated last across all subsystems together.

Component / Resource	Purpose
ESP32 microcontroller	Central processing, 12-bit ADC sampling and Wi-Fi/BLE IoT gateway
Dual servo motors	Azimuth and elevation actuation (0–180°, 50 Hz PWM)
LDR array	Optical differential tracking for alignment correction
DHT11 sensor	Temperature (0–50°C) and humidity (20–90% RH) sensing
HD44780 LCD with PCF8574 backpack	Local display of live azimuth/elevation and environmental readings
Power distribution rail (3.3 V / 5.0 V, 2 A peak)	Clean, regulated supply for logic and motor circuits

4.2 Hardware / Software Implementation & Modern Tools Used

Hardware: The azimuth base servo and elevation upper servo are mounted on a rigid gantry designed to resist wind shear and vibration; the LDR array is arranged in quadrants to allow differential intensity comparison; the DHT11 and LDR readings and computed angles are displayed locally on the two-row LCD (Row 1: “Azi: 91° Ele: 57°” style live angles; Row 2: “L:35% T:38C H:26%” environmental summary).

Software: A closed-loop state machine on the ESP32 continuously reads the 12-bit ADC values (4096 quantisation steps) from the sensors, compares the computed orientation against calibrated reference angles, and commands the servos only when the error exceeds a defined bound — avoiding unnecessary motor movement. Telemetry is published to the cloud platform for the dashboard, and override commands from the dashboard sliders are read back into the same control loop.

Tool / Software / Equipment	Purpose	Stage Used
Arduino IDE / ESP-IDF	Firmware development for ESP32	Embedded development
IoT cloud dashboard platform	Telemetry visualisation, historical logs, manual override	Integration & monitoring
[Multimeter / oscilloscope, if used]	Verifying PWM signal timing and rail voltages	Hardware bring-up & testing

[Insert prototype photo / assembled hardware screenshot / dashboard screenshot here.]

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Azimuth/elevation range	Command full sweep and observe physical angle with protractor/reference marker	Reaches 0° and 180° at both ends without mechanical binding
Angular resolution	Step command in 1° increments, verify servo response	Servo responds to each 1° step change
Closed-loop correction	Manually displace antenna and observe correction	System returns to reference angle without overshoot
Environmental sensing accuracy	Compare DHT11 reading against a reference thermometer/hygrometer	Within ±2°C of reference
Cloud telemetry & override	Trigger override slider on dashboard, observe physical response	Antenna moves to commanded angle within [X] seconds

Specification / Achieved Results

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Azimuth range	0–180°	[Record measured value]	[Pass/Fail]
Elevation range	0–180°	[Record measured value]	[Pass/Fail]
Angular resolution	1°	[Record measured value]	[Pass/Fail]
Temperature accuracy	±2°C	[Record measured value]	[Pass/Fail]
Override response time	[Define target]	[Record measured value]	[Pass/Fail]

4.4 Results

[Insert results tables / graphs / dashboard screenshots / simulation outputs here — e.g. measured alignment accuracy over repeated trials, response-time plots, and cloud dashboard telemetry screenshots. Present the actual data collected during testing; do not carry over the specification table above as if it were measured data.]

4.5 Data Analysis & Engineering Judgment

[Interpret the measured results here: discuss positioning accuracy/error, response time, sensor accuracy versus the DHT11 datasheet tolerance, and any observed deviations from the target specifications (e.g. backlash, overshoot, telemetry latency) together with their likely engineering causes.]

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Automated dual-axis control	[Link to Section 4.3/4.4 test evidence]	[State status]
Closed-loop alignment accuracy	[Link to Section 4.3/4.4 test evidence]	[State status]
Environmental compensation	[Link to Section 4.3/4.4 test evidence]	[State status]

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Reduced manual effort/error	[Qualitative comparison against manual baseline]	[State status]
Remote supervision via cloud	[Link to Section 4.3/4.4 test evidence]	[State status]

4.7 Strengths & Limitations

Strengths:

- Real-time closed-loop control and continuous environmental feedback via the ESP32.
- Low power consumption relative to PC-based or heavier microcontroller alternatives, suitable for solar/battery operation.
- Built-in Wi-Fi/Bluetooth removes the need for wired monitoring interfaces.
- Simple integration path for additional sensors/actuators (e.g. GPS, accelerometer) for future precision upgrades.

Limitations:

- LDR-based optical tracking depends on ambient light/beacon visibility and may be less reliable at night or in overcast conditions.
- Prototype scale has not been validated on a full-size, heavy commercial antenna structure.
- [Add any further limitations observed during your own testing, e.g. Wi-Fi range, servo torque limits, dashboard latency.]

4.8 Project Planning, Roles & Budget

[Insert Gantt chart / milestone timeline here.]

Student	Assigned Responsibility	Actual Contribution
Santhosh D (192412209)	[Assigned responsibility]	[Actual contribution]
A.K. Likith Charan (192412305)	[Assigned responsibility]	[Actual contribution]
Sanjay Anand (192412036)	[Assigned responsibility]	[Actual contribution]
Balakumar (192412221)	[Assigned responsibility]	[Actual contribution]
Mahendran (192412159)	[Assigned responsibility]	[Actual contribution]

Item	Estimated Cost	Actual Cost
ESP32 microcontroller	[₹]	[₹]
Servo motors (x2)	[₹]	[₹]
LDR array / sensors / DHT11	[₹]	[₹]
LCD display + backpack	[₹]	[₹]
Mechanical mount / enclosure	[₹]	[₹]

Item	Estimated Cost	Actual Cost
Power supply / regulators	[₹]	[₹]
Total	[₹]	[₹]

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The system successfully automates antenna positioning and alignment to ensure accurate orientation, eliminating the need for continuous manual adjustment. Signal strength is improved through continuous LDR-based monitoring and automatic corrective movement of the azimuth and elevation servos. Manual effort and alignment errors, inherent in conventional hand-adjusted mounts, are significantly reduced by the closed-loop control process. Integration with a cloud monitoring platform enables remote access and real-time status updates — including live positional angles, environmental readings and manual override — from any location via a smartphone or web browser. Overall, the combination of dual-axis actuation, closed-loop feedback, environmental compensation and cloud telemetry greatly enhances communication reliability and system efficiency compared with manual alignment approaches.

5.2 Future Work

- Incorporate GPS and magnetometer/accelerometer data for autonomous tracking of moving satellites or drones, rather than static-source alignment.
- Add solar-panel charging and battery management to make the system fully self-powered for remote deployments.
- Extend the cloud platform with predictive/AI-based misalignment detection using historical telemetry logs.
- Validate the design on a full-scale commercial antenna mount and under extended outdoor exposure testing.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- [List new technical knowledge gained, e.g. closed-loop control design, PWM servo driving, IoT dashboard integration.]

Engineering & Professional Skills Developed:

- [List skills developed, e.g. requirements analysis, trade-off evaluation, teamwork, technical documentation.]

Individual Reflection (each student, 4–5 lines):

Santhosh D (192412209):

[Most difficult engineering problem encountered and how it was solved; a key engineering decision made personally and the evidence behind it; new knowledge acquired independently; what would be redesigned if repeated.]

A.K. Likith Charan (192412305):

[Individual reflection — 4–5 lines.]

Sanjay Anand (192412036):

[Individual reflection — 4–5 lines.]

Balakumar (192412221):

[Individual reflection — 4–5 lines.]

Mahendran (192412159):

[Individual reflection — 4–5 lines.]

REFERENCES

Use a consistent recognised referencing style such as IEEE. All literature, standards, websites, datasets, software, open-source resources and AI-assisted resources used must be appropriately acknowledged. Example entries to complete/replace with your actual sources:

[1] [Author(s)], “[Paper/Article title],” [Journal/Conference], vol. X, no. X, pp. XX–XX, [Year].

[2] Espressif Systems, “ESP32 Series Datasheet,” [online]. Available: [URL]. [Accessed: date].

[3] [DHT11 sensor datasheet / manufacturer, URL, accessed date].

[Add all literature, standards, datasheets, and AI-assisted tools (e.g. Claude) used during the project, per Chapter 2 and the Declaration.]

APPENDICES

Appendix A – Detailed Calculations

[Insert PWM duty-cycle calculations, ADC quantisation calculations, power-budget calculations, etc.]

Appendix B – Engineering Drawings / Schematics

[Insert circuit schematic and mechanical mount drawings.]

Appendix C – Source Code / Code Repository Information

[State repository link/branch; include only essential code excerpts here.]

Appendix D – Datasheets

[Attach/summarise ESP32, servo, LDR and DHT11 datasheets.]

Appendix E – Experimental / Raw Data

[Insert raw test logs.]

Appendix F – Ethics / Institutional Approval (where applicable)

[Attach if applicable.]

Appendix G – Project Meeting / Progress Records

[Insert meeting minutes / progress log.]

Appendix H – User/Industry Feedback (where applicable)

[Attach if applicable.]

Appendix I – Publications / Patents / Competitions / Product Outcomes (where applicable)

[Attach if applicable.]

Appendix J – Individual Contribution Evidence (mandatory for team projects)

[Attach supporting evidence for the Individual Contribution Statement.]

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)